

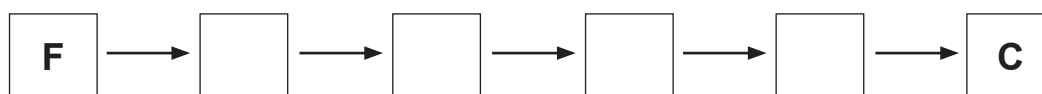
5. (a) The first stage in the preparation of a salt from an acid and an alkali often involves carrying out a titration.



A-F below are the steps of the method that can be used to prepare a salt from an acid and an alkali. However, these steps are **not** in the correct order.

- A** Repeat without the indicator using the same volumes of alkali and acid
- B** Add 2-3 drops of indicator
- C** Allow the solution to cool and crystallise naturally
- D** Place the salt solution into an evaporating dish and heat until half the volume of solution remains
- E** Add the acid to the alkali until the indicator changes colour
- F** Measure 25.0 cm³ of alkali into a conical flask

- (i) Place the steps of the method into the correct order. The first and last steps have already been added. [2]



- (ii) Give the **letter** of the step where error is **most** likely to occur. Give the reason for your choice. [2]

Step

Reason

.....



- (b) The reaction between an acid and an alkali produces a salt and water. The following equation shows how potassium sulfate is produced from an acid and an alkali.

potassium hydroxide +[?]..... $\longrightarrow$ potassium sulfate + water

- (i) Give the name of the missing acid.

[1]

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- (ii) Give the chemical formula of potassium sulfate.

[1]

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